

Control of T cell antigen reactivity via programmed TCR downregulation

Alena M Gallegos^{1,3,4}, Huizhong Xiong^{1,4}, Ingrid M Leiner¹, Bože Sušac¹, Michael S Glickman^{1,2}, Eric G Pamer^{1,2} & Jeroen W J van Heijst^{1,3}

The T cell antigen receptor (TCR) is unique in that its affinity for ligand is unknown before encounter and can vary by orders of magnitude. How the immune system regulates individual T cells that display very different reactivity to antigen remains unclear. Here we found that activated CD4⁺ T cells, at the peak of clonal expansion, persistently downregulated their TCR expression in proportion to the strength of the initial antigen recognition. This programmed response increased the threshold for cytokine production and recall proliferation in a clone-specific manner and ultimately excluded clones with the highest antigen reactivity. Thus, programmed downregulation of TCR expression represents a negative feedback mechanism for constraining T cell effector function with a suitable time delay to thereby allow pathogen control while avoiding excess inflammatory damage.

Efficient recognition and control of microbial infections relies on the availability of a diverse T cell antigen receptor (TCR) repertoire^{1,2}. In developing T cells, TCR diversity is generated via somatic recombination, during which different gene segments are combined with the addition and removal of nucleotides at recombination junctions. This process results in TCRs with highly variable sequences and therefore generates individual T cells that recognize peptide ligands with a wide range of binding strengths (avidities). This has important functional consequences, as high-avidity T cells are generally more sensitive to antigen, undergo more proliferation during immune responses and produce more cytokines^{3–9}. However, T cells at the upper end of the avidity spectrum can also display substantial limitations in biological activity^{10–13}, which suggests the existence of regulatory mechanisms that prevent deleterious consequences of very strong antigen reactivity. Hence, the optimal avidity associated with long-term T cell functionality remains elusive, particularly in the context of immunity to microbial infection.

Mycobacterium tuberculosis remains one of the most detrimental human pathogens and has been challenging to combat due to the lack of an effective vaccine^{14,15}. Although CD4⁺ T cells are essential for the control of *M. tuberculosis*^{16,17}, bacterial eradication is rarely achieved. One explanation is that *M. tuberculosis* is able to inhibit major histocompatibility complex (MHC) class II expression^{18,19}, which results in limited antigen presentation in mycobacterial granulomas^{20,21}. Therefore, one way to improve vaccination against tuberculosis might be to engage T cells of an overall very high avidity, because these would require less antigen to activate effector mechanisms.

To investigate how T cell avidity influences immunity to *M. tuberculosis*, we generated a mouse (C24) in which CD4⁺ T cells

have transgenic expression of a TCR with very high avidity for the mycobacterial antigen ESAT6(1–20). By comparing this mouse with a previously published mouse (C7) that expresses a TCR with intermediate avidity for ESAT6 (refs. 22,23), we found that the T cells of higher avidity were less able to control bacterial growth, a phenotype that was associated with a substantial loss of TCR expression. On the basis of those results, we found that clonally expanded T cells underwent a program of persistent TCR downregulation that was proportional to the strength of initial antigen recognition. This previously unappreciated feedback mechanism increased the threshold for cytokine production and recall proliferation in an avidity-dependent manner, with the result that T cells of intermediate avidity made the greatest contribution to antimicrobial defense.

RESULTS

Avidity of C7 and C24 CD4⁺ T cells

For the investigation of CD4⁺ T cell immunity to tuberculosis, we used the previously generated C7 mouse, in which all CD4⁺ T cells have transgenic expression of a TCR specific for the mycobacterial antigen ESAT6(1–20). Studies of this mouse have demonstrated that C7 cells differentiated into the T_H1 subset of CD4⁺ helper T cells are able to greatly diminish but not eliminate mycobacterial growth^{22,23}. To determine whether that failure of bacterial eradication is a general feature of T cell immunity, we generated another mouse with transgenic TCR expression, using a ESAT6(1–20)-specific T cell hybridoma (C24) obtained from an *M. tuberculosis*-infected C57BL/6J mouse. We chose the C24 TCR because it uses α - and β -chains different from those used by the C7 TCR, which created the possibility

¹Immunology Program, Sloan-Kettering Institute, Memorial Sloan-Kettering Cancer Center, New York, New York, USA. ²Infectious Diseases Service, Department of Medicine, Memorial Sloan-Kettering Cancer Center, New York, New York, USA. ³Present addresses: National Institute for Research in Tuberculosis, Chennai, India (A.M.G.), and Center of Experimental and Molecular Medicine, Academic Medical Center, University of Amsterdam, Amsterdam, the Netherlands (J.W.J.v.H.).

⁴These authors contributed equally to this work. Correspondence should be addressed to J.W.J.v.H. (j.w.vanheijst@amc.uva.nl).

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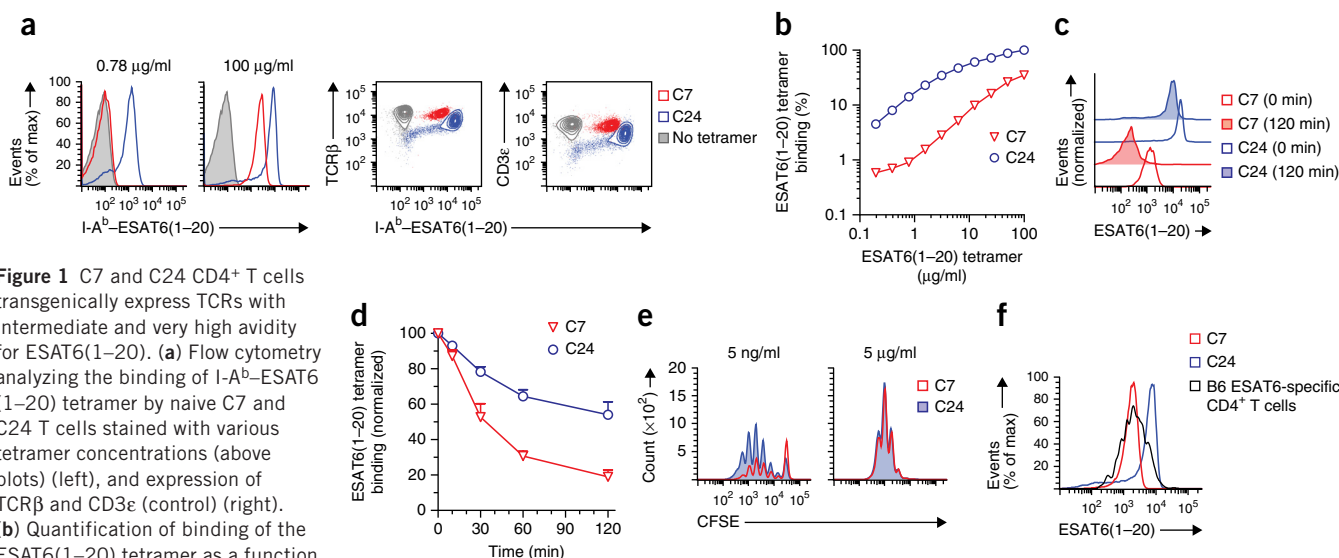


Figure 1 C7 and C24 CD4⁺ T cells transgenically express TCRs with intermediate and very high avidity for ESAT6(1–20). **(a)** Flow cytometry analyzing the binding of I-A^b-ESAT6(1–20) tetramer by naive C7 and C24 T cells stained with various tetramer concentrations (above plots) (left), and expression of TCR β and CD3 ϵ (control) (right). **(b)** Quantification of binding of the ESAT6(1–20) tetramer as a function of tetramer concentration, with tetramer binding assessed as median fluorescence intensity and normalized to the highest signal obtained for C24 T cells. **(c)** Flow cytometry analyzing dissociation of the ESAT6(1–20) tetramer by naive C7 and C24 T cells stained with tetramer at a concentration of 50 μ g/ml; results are normalized to the mode. **(d)** Dissociation of the ESAT6(1–20) tetramer as a function of time; results are normalized to those at time 0, set as 100. **(e)** Flow cytometry analyzing dilution of the cytosolic dye CFSE by C7 and C24 T cells activated for 4 d *in vitro* with various concentrations (above plots) of ESAT6(1–20) peptide. **(f)** Flow cytometry analyzing binding of the ESAT6(1–20) tetramer by C7 and C24 T cells, and endogenous CD4⁺ T cells (B6 ESAT6-specific CD4⁺ T cells), isolated from the lungs of a C57BL/6J mouse 28 d after infection with *M. tuberculosis* and then stained with tetramer at a concentration of 50 μ g/ml. Data are from one experiment representative of two independent experiments (mean $\pm$ s.d. of $n = 3$ male mice per group (d) or $n = 5$ male mice per group (f)).

that it might recognize ESAT6(1–20) with a different affinity. To test this, we incubated naive C7 and C24 T cells with various concentrations of ESAT6(1–20) tetramers, to evaluate their antigen-binding strength²⁴. At saturating concentrations, C24 T cells bound threefold more tetramer than did C7 T cells, and this difference increased to 15-fold at non-saturating concentrations (Fig. 1a,b). Since the TCR expression of naive C7 T cells and naive C24 T cells was comparable (Fig. 1a), this suggested that C24 T cells had substantially greater structural avidity for ESAT6(1–20), a conclusion confirmed by tetramer-dissociation studies (Fig. 1c,d). To investigate whether C7 T cells and C24 T cells also differed in functional avidity, we activated these cells with various concentrations of ESAT6(1–20) peptide. At a low antigen dose, 45% of C24 T cells had proliferated after 4 d, compared with only 18% of C7 T cells, whereas at a high antigen dose, there was no difference (Fig. 1e), which indicated that the degree to which the C24 TCR was stimulated was higher. Consistent with that, activated C24 T cells also displayed increased expression of the immunomodulatory receptor CTLA-4 (CD152) than that of activated C7 T cells²⁵ (Supplementary Fig. 1).

Because both C7 T cells and C24 T cells were generated with TCRs of *M. tuberculosis*-infected mice, we sought to compare their avidity with that of ESAT6(1–20)-specific endogenous CD4⁺ T cells. Therefore, we infected C57BL/6J mice with *M. tuberculosis* and isolated ESAT6(1–20) tetramer-binding cells at the peak of the response. We observed a normal distribution of tetramer binding by endogenous CD4⁺ T cells that spanned two orders of magnitude, which probably reflected cells with different avidities for ESAT6(1–20) (Fig. 1f). Notably, ~50% of tetramer-positive endogenous CD4⁺ T cells bound tetramer as well as or better than C7 T cells did, whereas only ~5% bound tetramer as well as or better than C24 T cells did. This suggested that, relative to endogenous CD4⁺ T cells, C7 T cells and C24 T cells represented T cells with intermediate avidity and very high avidity, respectively.

Control of *M. tuberculosis* by C7 and C24 T cells

Given the very high avidity of C24 T cells, we anticipated that these cells might be very efficient at controlling *M. tuberculosis* infection, which is associated with limited antigen presentation^{20,21}. Since activated T_H1 cells confer greater protection than do naive T cells²³, we polarized C7 and C24 T cells *in vitro* using T_H1 conditions, transferred these cells into mice and infected the host mice with *M. tuberculosis*. Analysis of peripheral blood revealed that C7 and C24 T_H1 cells continued to expand for 10 d after initial activation (Fig. 2a). The expansion of both clones peaked at a similar frequency (Fig. 2a), which indicated that their capacity for proliferation was not inherently different. To assess bacterial control, we harvested lungs on day 16 after infection, when protection by transferred T_H1 cells is known to be maximal²². Notably, although both T_H1 clones had inhibited bacterial growth, the degree of protection by C24 T_H1 cells was significantly lower than that achieved by C7 T_H1 cells (Fig. 2b), indicative of impaired functionality associated with very high antigen avidity.

To investigate that apparent failure of C24 T cells, we analyzed the expression of several activation markers. Although C24 T_H1 cells from the lungs were uniformly CD44^{hi}, they displayed substantial downregulation of TCR expression (Fig. 2c). Specifically, the TCR expression of C24 T_H1 cells was only 6% that of endogenous CD4⁺ T cells, whereas the TCR expression of C7 T_H1 cells was 78% that of endogenous CD4⁺ T cells (Fig. 2d). Thus, the unexpected diminished performance of C24 T_H1 cells could have been due to loss of TCR expression.

Programmed TCR downregulation by C7 and C24 T cells

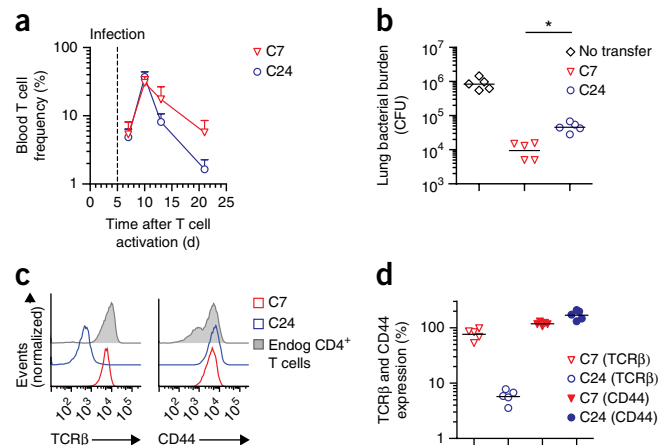
In mice, *M. tuberculosis* causes a chronic infection during which ESAT6 is persistently expressed²⁶. Hence, we reasoned that the downregulation of TCR expression by C24 T_H1 cells could have been driven by chronic antigen exposure. To test this, we transferred C7 or C24 T_H1 cells into mice that we then either infected with *M. tuberculosis*

Figure 2 Differential capacity of C7 T_H1 cells and C24 T_H1 cells to control *M. tuberculosis* infection. (a) Frequency of C7 or C24 T cells among $CD4^+$ T cells in the blood of mice given 5×10^6 CD90.1 $^+$ C7 or C24 T_H1 cells that had been activated *in vitro* for 4 d, followed by infection of host mice with *M. tuberculosis*. (b) Bacterial burden in the lungs of recipient mice in a, as well as in mice that did not receive any T cells (No transfer), assessed at 16 d after infection and presented as colony-forming units (CFU). (c,d) Flow cytometry analyzing the expression of TCR β and CD44 by C7 and C24 T_H1 cells, and by endogenous ((CD90.1 $^-$) $CD4^+$ T cells (Endog $CD4^+$ T cells), from the lungs of recipient mice as in b (results normalized to the mode) (c), and aggregate data obtained as in c, assessed as median fluorescence intensity and normalized to the signal obtained for endogenous $CD4^+$ T cells (d). Each symbol (b,d) represents an individual mouse; small horizontal lines indicate the geometric mean. * $P < 0.001$ (unpaired two-tailed Student's *t* test). Data are from one experiment representative of three independent experiments (mean $\pm$ s.d. of $n = 5$ female mice per group in a).

or left uninfected. As before, C7 and C24 T_H1 cells continued to proliferate for several days after adoptive transfer, then underwent a contraction in cell number (Fig. 3a). The extent of T_H1 cell clonal expansion and contraction was very similar for *M. tuberculosis*-infected mice and uninfected mice (Fig. 3a), which indicated that these properties were programmed during initial T cell activation *in vitro*^{27–31}. Notably, both C7 T_H1 cells and C24 T_H1 cells exhibited high TCR expression during the expansion phase (Fig. 3b,c). However, at the peak of clonal expansion, both clones had downregulated their TCR expression, and this downregulation persisted throughout the contraction phase (Fig. 3b,c). By day 16 after activation, the TCR expression of C7 T_H1 cells and C24 T_H1 cells from the blood was only 28% and 4%, respectively, that of endogenous $CD4^+$ T cells (Fig. 3b,c). Therefore, both T_H1 clones had downregulated their TCR expression, but they had done so to different degrees, with the clone of higher avidity displaying more TCR downregulation. Unexpectedly, the extent of TCR downregulation was the same for infected mice and uninfected mice (Fig. 3c), which suggested that this process was not driven by chronic antigen exposure but was instead programmed during initial T cell activation.

Although the substantial downregulation of TCR expression by C24 T_H1 cells did not require continuous exposure to foreign antigen, it could still have resulted from greater self-reactivity. Naive C24 T cells displayed 50% lower expression of the negative regulator CD5, but 70% greater basal phosphorylation of TCR ζ , than that of naive C7 T cells (Supplementary Fig. 2), which suggested that of the two sets of T cells, C24 T cells had somewhat higher self-reactivity. However, transfer of C7 or C24 T_H1 cells into wild-type or MHC class II-deficient mice revealed indistinguishable downregulation of TCR expression (Supplementary Fig. 3), which indicated that no further TCR signaling was needed to induce this downregulation beyond the initial T cell activation and thus the response appeared to be truly programmed.

Figure 3 Clonally expanded C7 and C24 T_H1 cells undergo persistent TCR downregulation that is programmed during initial T cell activation. (a) Frequency of C7 or C24 T cells among $CD4^+$ T cells in the blood of mice given 1×10^6 CD90.1 $^+$ C7 or C24 T_H1 cells that had been activated *in vitro* for 3 d, followed by infection of host mice with *M. tuberculosis* (inf) or no infection (left uninfected (UI)). (b,c) Flow cytometry analyzing the kinetics of TCR β expression by C7 and C24 T_H1 cells, and endogenous (CD90.1 $^-$) $CD4^+$ T cells, from the blood of recipient mice in a, assessed on days 6 and 16 after infection (above plots) (results normalized to the mode) (b), and analysis over time of data obtained as in b, presented relative to results obtained for the endogenous $CD4^+$ T cells (c). Data are from one experiment representative of two independent experiments (mean $\pm$ s.d. of $n = 3$ male mice per group in a,c).



Programmed TCR downregulation by endogenous $CD4^+$ T cells

The finding that both C7 T_H1 cells and C24 T_H1 cells displayed downregulation of TCR expression raised the question of whether this represented a general trait of activated $CD4^+$ T cells. To investigate this *in vivo*, we generated a recombinant *Listeria monocytogenes* strain that expresses ESAT6 (*L. monocytogenes*-ESAT6)³². *L. monocytogenes* induces potent T cell responses to multiple antigens and therefore we anticipated that use of the ESAT6-expressing strain would facilitate the activation of naive C7 and C24 T cells, as well as that of endogenous *L. monocytogenes*-specific $CD4^+$ T cells *in vivo*. To allow unbiased assessment of TCR downregulation on endogenous *L. monocytogenes*-specific T cells, we reasoned that flow cytometry with gating on activation markers might enable the identification of such cells without requiring TCR-based methodology. To test this, we transferred naive CD90.1 $^+$ C7 or C24 T cells into mice and infected the mice with *L. monocytogenes*-ESAT6. A third group of infected mice did not receive any cells ('no transfer'), to allow natural development of the endogenous *L. monocytogenes*-specific response. On day 7 after infection, C7 and C24 T cells from the blood were readily identified by their expression of the alloantigen CD90.1, and further gating revealed that these cells uniformly had an activated phenotype, being $CD44^{hi}CD62L^-$ (Fig. 4a). In contrast, gating on endogenous $CD4^+$ T cells from 'no-transfer' recipients revealed two subpopulations, with cells having either an activated phenotype or a naive ($CD44^{lo}CD62L^+$) phenotype.

We then assessed the response kinetics of activated C7 and C24 T cells, as well as that of endogenous $CD44^{hi}CD62L^-$ $CD4^+$ T cells.

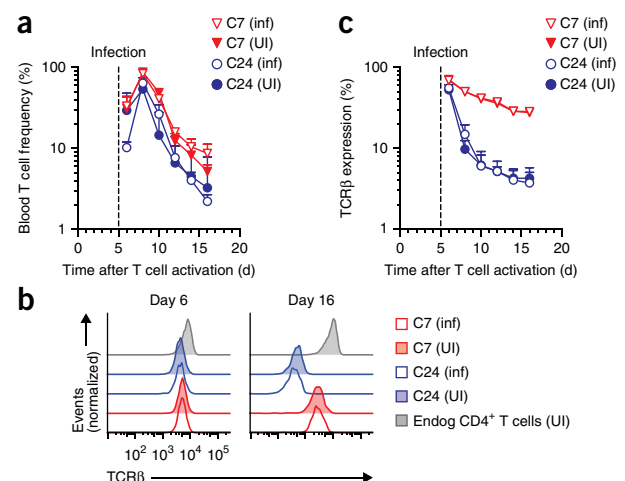
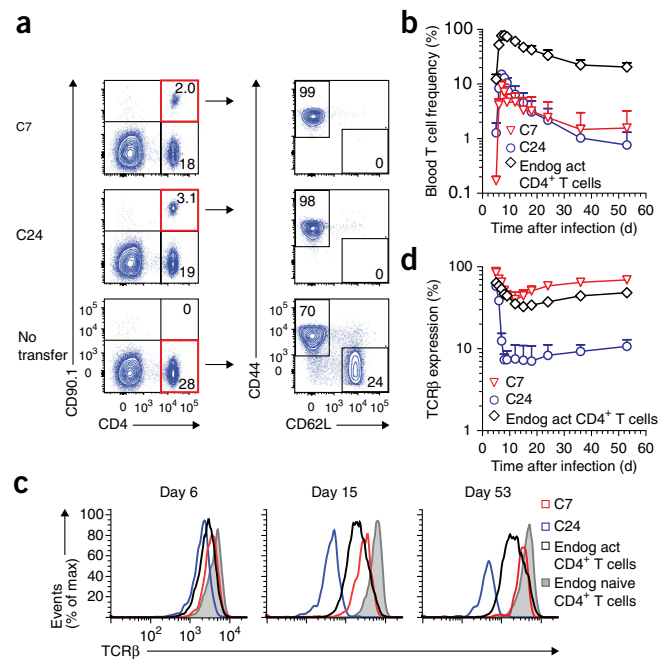


Figure 4 Programmed TCR downregulation is a general feature of activated CD4⁺ T cells. **(a)** Flow cytometry analyzing the expression of CD90.1 and CD4 by T cells (left), and of CD44 and CD62L by the CD4⁺ T cells gated at left (right), in blood from mice given 1×10^4 naive CD90.1⁺ C7 or C24 CD4⁺ T cells or no cells (No transfer), then infected with recombinant *L. monocytogenes*–ESAT6, analyzed on day 7 after infection. Numbers in quadrants (left) indicate percent CD90.1⁺ (donor) CD4⁺ T cells (top) or CD90.1[−] (host) T cells (bottom); numbers in outlined areas (right) indicate percent CD44^{hi}CD62L[−] (activated) T cells (top left) or CD44^{lo}CD62L⁺ (naive) T cells (bottom right) among CD4⁺ T cells. **(b)** Frequency of C7 and C24 T cells among CD4⁺ T cells in blood from the recipient mice in **a**, as well as of endogenous activated (CD44^{hi}CD62L[−]) CD4⁺ T cells in blood from mice in **a** that did not receive donor cells (Endog act CD4⁺ T cells). **(c,d)** Flow cytometry analyzing the kinetics of TCR β expression by C7 and C24 CD4⁺ T cells in blood from the recipient mice in **a**, as well as endogenous activated (CD44^{hi}CD62L[−]) and naive (CD44^{lo}CD62L⁺) CD4⁺ T cells in blood from mice in **a** that did not receive donor cells, assessed on days 6, 15 and 53 after infection (above plots) **(c)**, and analysis over time of the data obtained as in **b**, presented relative to results obtained for the endogenous naive CD4⁺ T cells **(d)**. Data are from one experiment representative of two independent experiments (mean + s.d. of $n = 5$ male mice per group in **b,d**).

As expected, C7 and C24 T cells proliferated until day 7 after infection, after which the response contracted, leaving a memory T cell population by day 35 (**Fig. 4b**). Notably, the expansion and contraction pattern of endogenous CD44^{hi}CD62L[−] CD4⁺ T cells was very similar to that of C7 and C24 T cells, which suggested that this gating strategy identified activated *L. monocytogenes*–specific T cells.

Being able to detect endogenous activated CD4⁺ T cells independently of their TCR, we investigated the kinetics of TCR expression. Similar to our results obtained with *in vitro*–activated T_H1 cells, *in vivo*–activated C7 and C24 T cells exhibited high TCR expression during initial proliferation, with expression rapidly downregulated at the peak of clonal expansion (**Fig. 4c,d**). The TCR expression reached its lowest amount 15 d after infection, after which expression increased somewhat (**Fig. 4c,d**). However, as late as 53 d after infection, the TCR expression of C24 T cells was still only 11% that of endogenous naive CD4⁺ T cells (**Fig. 4c,d**), which suggested that once the adjusted TCR expression level was set, it remained very stable. Despite having almost no TCR expression, clonally expanded



C24 T cells had normal expression of costimulatory molecules, activation markers, cytokine receptors and inhibitory receptors and did not express the transcription factor Foxp3 (**Supplementary Fig. 4**), which indicated that they were ‘ordinary’ effector cells. Notably, endogenous activated CD4⁺ T cells also displayed persistent downregulation of TCR expression, with kinetics similar to those of C7 and C24 T cells (**Fig. 4c,d**). Although the median downregulation of TCR expression by endogenous T cells resembled that of C7 T cells, there were clearly cells present in this population that had TCR expression equivalent to that of C24 T cells (**Fig. 4c,d**).

To investigate whether programmed TCR downregulation also occurred during chronic bacterial infection, we analyzed the responses of naive C7 and C24 T cells activated by *M. tuberculosis* *in vivo*; this revealed the same pattern of downregulation of TCR expression as

Figure 5 Programmed TCR downregulation is associated with degradation of TCR ζ .

(a) Frequency of C7 or C24 T cells or endogenous activated CD4⁺ T cells (defined as in **Fig. 4b**) among CD4⁺ T cells in blood from mice given 1×10^4 naive CD90.1⁺ C7 or C24 CD4⁺ T cells or no cells, then infected with recombinant *L. monocytogenes*–ESAT6. **(b–d)** Flow cytometry analyzing the kinetics of the surface expression of TCR β (top) and total expression of TCR ζ (bottom) in C7 and C24 CD4⁺ T cells, as well as endogenous activated and naive CD4⁺ T cells (defined as in **Fig. 4c**), in blood from the recipient mice in **a**, assessed on days 5, 14 and 38 after infection (above plots) **(b)**, and analysis over time of the data obtained for TCR β **(c)** and TCR ζ **(d)** as in **b**, presented relative to results obtained for the endogenous naive CD4⁺ T cells **(c,d)**. **(e)** Pearson correlation of surface expression of TCR β and total expression of TCR ζ by C7 and C24 CD4⁺ T cells, as well as endogenous activated CD4⁺ T cells ($r = 0.86$). Data are from one experiment representative of two independent experiments (mean + s.d. of $n = 5$ male mice per group **(a,c,d)** or mean of $n = 5$ per group **(e)**).

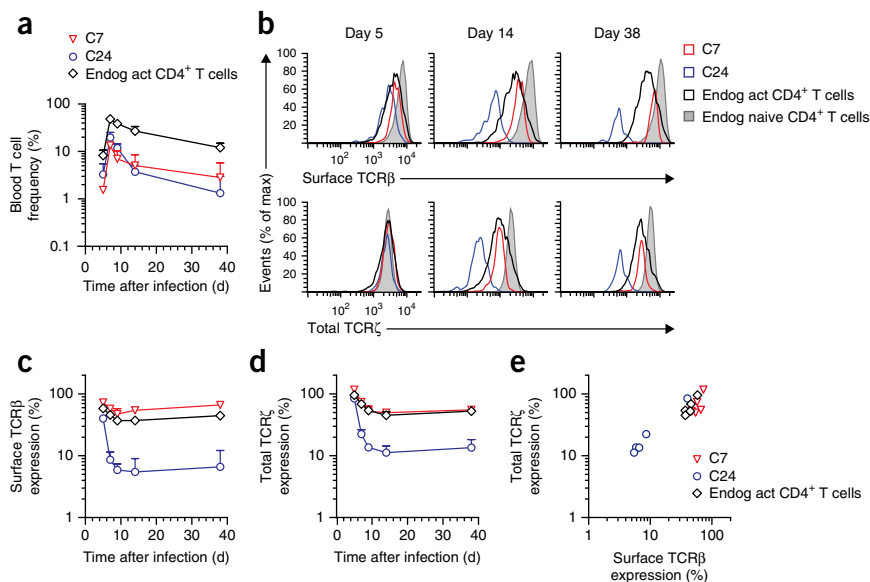
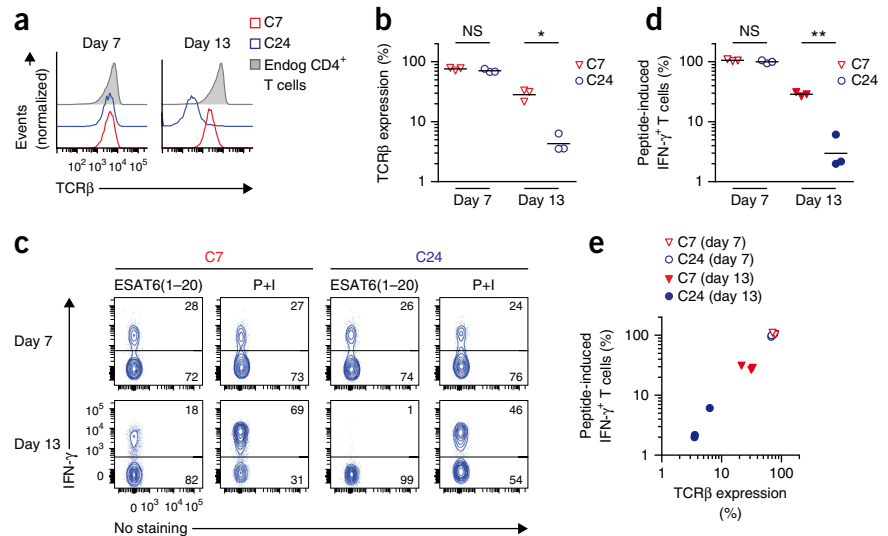


Figure 6 Programmed TCR downregulation controls CD4⁺ T cell cytokine production. (a,b) Flow cytometry analyzing the TCR β expression of C7 or C24 T cells or endogenous (CD90.1⁻) CD4⁺ T cells among splenocytes from mice given 1×10^6 CD90.1⁺ C7 or C24 T_H1 cells that had been activated *in vitro* for 4 d, assessed on day 7 or 13 after activation (above plots) (results normalized to the mode) (a), and aggregate data obtained as in a, presented relative to results obtained for endogenous CD4⁺ T cells (b). (c,d) Flow cytometry analyzing the expression of IFN- γ by splenic C7 and C24 T_H1 cells obtained from the recipient mice in a, day 7 or 13 after activation, then restimulated with ESAT6(1–20) peptide or PMA plus ionomycin (P+I) (c), and aggregate data obtained as in c, presented relative to results obtained for cells stimulated with PMA plus ionomycin (d). Numbers in outlined areas (c) indicate percent IFN- γ ⁺ cells (top) or IFN- γ ⁺ cells (bottom). (e) Pearson correlation of TCR β expression (presented relative to results obtained for endogenous CD4⁺ T cells) and TCR-dependent production of IFN- γ (presented relative to results obtained for cells stimulated with PMA plus ionomycin) by C7 and C24 T_H1 cells as in a, s ($r = 0.98$). Each symbol (b,d,e) represents an individual mouse; small horizontal lines indicate the geometric mean. NS, not significant ($P > 0.05$); * $P = 0.003$ and ** $P < 0.001$ (unpaired two-tailed Student's *t* test). Data are from one experiment representative of two independent experiments.



the downregulation during acute infection with *L. monocytogenes* (Supplementary Fig. 5). Furthermore, endogenous *M. tuberculosis*-specific CD4⁺ T cells also displayed downregulation of TCR expression, and cells that weakly bound the ESAT6(1–20) tetramer displayed lower TCR expression than that of cells that strongly bound the tetramer (Supplementary Fig. 5), which indicated that tetramer analyses can be confounded by TCR downregulation. Together these data established that activated CD4⁺ T cells displayed a program of persistent downregulation of TCR expression that was initiated at the peak of clonal expansion and resulted in a repertoire with 'graded' TCR expression, of which C7 T cells and C24 T cells represented an average example and an extreme example, respectively.

Programmed TCR downregulation linked to TCR ζ degradation

Given that the α - and β -chains of C7 and C24 TCRs were driven by a constitutive promoter, we hypothesized that the downregulation of TCR expression was regulated by other components of the TCR complex. Activated C24 T cells with almost completely absent surface TCR expression had unaltered transcription of the TCR complex components CD3 γ , CD3 δ , CD3 ϵ and TCR ζ (Supplementary Fig. 6), which suggested that downregulation of the TCR was mediated via post-transcriptional mechanisms. Assembly of the TCR complex is generally limited by the availability of TCR ζ ³³, and transient downregulation of TCR surface expression can be mediated via degradation of TCR ζ ^{34–36}. This prompted us to investigate the expression of TCR ζ protein in CD4⁺ T cells activated by *L. monocytogenes*. As before, C7 and C24 T cells, as well as endogenous activated CD4⁺ T cells, displayed graded and persistent downregulation of TCR expression, starting at the peak of clonal expansion (Fig. 5a–d). Notably, total expression of TCR ζ demonstrated a pattern of downregulation markedly similar to that of surface expression of TCR β (Fig. 5e), which suggested that programmed TCR downregulation was mediated via degradation of TCR ζ .

Programmed TCR downregulation controls antigen reactivity

TCR expression is critical for an efficient response to antigenic stimulation^{37–39}. To investigate to what extent programmed

TCR downregulation influences the antigen reactivity of activated T cells, we transferred C7 or C24 T_H1 cells into uninfected mice. This generated a setting in which both T_H1 clones had high TCR expression initially (day 7 after activation), whereas later (on day 13), TCR expression was intermediate on C7 T_H1 cells and low on C24 T_H1 cells (Fig. 6a,b). To functionally test these cells, we restimulated them with ESAT6(1–20) peptide, to measure the TCR-dependent capacity for cytokine production. As a control, we incubated the same cells with the phorbol ester PMA plus ionomycin, which allows measurement of the TCR-independent capacity for cytokine production. On day 7, TCR-dependent production of interferon- γ (IFN- γ) by C7 and C24 T_H1 cells was equal to their TCR-independent production of IFN- γ (Fig. 6c,d), which indicated that when TCR expression was high, the antigen reactivity of both clones was fully intact. In contrast, on day 13, the TCR-dependent production of IFN- γ by C7 T_H1 cells was intermediate and that of C24 T_H1 cells was low, which resulted in a very strong correlation between TCR expression and TCR-dependent production of IFN- γ (Fig. 6c–e). Notably, the TCR-independent production of IFN- γ by both clones was very high on day 13 (Fig. 6c), which indicated that following TCR downregulation, the T cells continued to have a large intrinsic capacity to produce cytokines; however, their cytokine production in response to antigen was drastically altered.

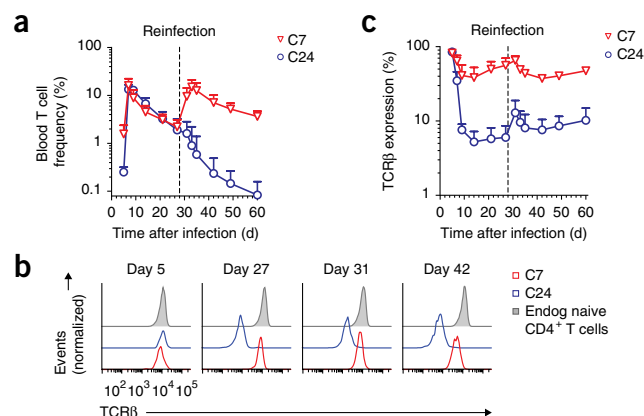
Since programmed TCR downregulation was initiated at the peak of clonal expansion and persisted for at least 50 d, we reasoned this might have a substantial effect on participation in secondary immune responses. To address this, we infected recipients of naive C7 or C24 T cells with *L. monocytogenes*–ESAT6 and reinfected these mice 1 month later. As before, the primary clonal expansion of C7 and C24 T cells was similar, and TCR expression on both clones was initially high, whereas by day 27, it was intermediate on C7 T cells and very low on C24 T cells (Fig. 7). Strikingly, rechallenge on day 28 resulted in an increase in the frequency of C7 T cells of >500% by day 33, whereas there was a decrease of 50% in the frequency of C24 T cells, which indicated that these cells were not able to participate in the secondary response.

Figure 7 Programmed TCR downregulation controls CD4⁺ T cell recall proliferation. **(a)** Frequency of C7 or C24 T cells among CD4⁺ T cells in blood from mice given 1×10^4 naive CD90.1⁺ C7 or C24 CD4⁺ T cells, then infected with recombinant *L. monocytogenes*-ESAT6 on day 0 and reinfected with the same bacteria (dashed vertical line) on day 28. **(b,c)** Flow cytometry analyzing the kinetics of TCR β expression by C7 and C24 CD4⁺ T cells, as well as endogenous naive (CD44^{lo}CD62L⁺) CD4⁺ T cells, in blood from the recipient mice in **a**, assessed on days 5, 27, 31 and 42 after infection (above plots) (results normalized to the mode) **(b)**, and analysis over time of the data obtained as in **b**, relative to results obtained for the endogenous naive CD4⁺ T cells **(c)**. Data are from one experiment representative of three independent experiments (mean + s.d. of $n = 5$ male mice per group in **a,c**).

As *L. monocytogenes* induces the clonal expansion of both CD4⁺ T cells and CD8⁺ T cells, we sought to determine whether CD8⁺ T cells would also undergo downregulation of TCR expression. To this end, we gated on endogenous activated CD8⁺ T cells (CD44^{hi}CD62L⁻) after infection of mice with *L. monocytogenes* and found that the gated cells displayed persistent downregulation of TCR expression with kinetics similar to those of activated CD4⁺ T cells (**Supplementary Fig. 7**). Therefore, programmed TCR downregulation appeared to be a feature of both T cell lineages for the regulation of antigen reactivity after clonal expansion.

Programmed TCR downregulation is driven by T cell avidity

Given that activated C7 T cells and activated C24 T cells displayed TCR downregulation to a different degree, we wondered whether this was a reflection of their differential avidity for antigen. Activation of C24 T cells with a peptide dose 99% lower than the standard dose used above, which resulted in greatly diminished clonal expansion, had only a minimal effect on the extent to which TCR expression was downregulated (**Supplementary Fig. 8**), in agreement with published studies showing that small quantities of peptide can still engage many TCRs via serial triggering^{40,41}. Therefore, to truly manipulate T cell avidity, we screened mutants of the ESAT6(1–20) peptide and identified one such mutant, F8A, that was a full agonist for C7 T cells but only a partial agonist for C24 T cells (**Fig. 8a**). We then generated C24 T_H1 cells in the presence of wild-type ESAT6(1–20) or the F8A mutant peptide and adoptively transferred these cells into mice. Notably, following clonal expansion, C24 T_H1 cells activated with wild-type ESAT6(1–20) displayed 80% greater downregulation of TCR expression than that of C24 T_H1 cells activated with the F8A mutant,



and this difference was maintained for up to 1 month (**Fig. 8b,c**). Together these data indicated that T cell avidity was a driving factor behind the extent of TCR downregulation, which suggested that the clonally expanded T cells adjusted their reactivity according to the strength of the initial antigen recognition.

DISCUSSION

Among cell-surface receptors, the TCR is unique in that its affinity for ligand is not fixed but is highly variable. Therefore, how strongly an individual T cell will react to an antigen is essentially unknown before its encounter with that antigen. Given that interactions of higher avidity generally lead to greater biological output, this raises the question of whether negative feedback mechanisms exist that regulate T cell effector function in an avidity-dependent manner. In this study, we have identified such a mechanism and have shown that it operated via programmed downregulation of TCR expression following clonal expansion and thereby effectively lowered the functional avidity of the TCR-ligand interaction.

What is the rationale for programmed TCR downregulation and why does it lead to greater attenuation of high-avidity T cells than that of intermediate-avidity T cells? One explanation is that greater suppression of high-avidity cells preserves the polyclonality of the response and thereby counteracts a pathogen's escape from the immune system^{13,25}. In general, the aim of any response is to control pathogen replication with the least amount of collateral damage. Programmed TCR downregulation in that sense represents a typical example of 'reactive regulation', which means a negative-feedback response that is executed proportional to the strength of signaling input but with a suitable time delay⁴². This enables control of the pathogen while the response is automatically attenuated to avoid excess damage to the host. Given the reactive nature of programmed TCR downregulation, this indicates that at the upper end of the avidity spectrum, cells will be excluded from further participation

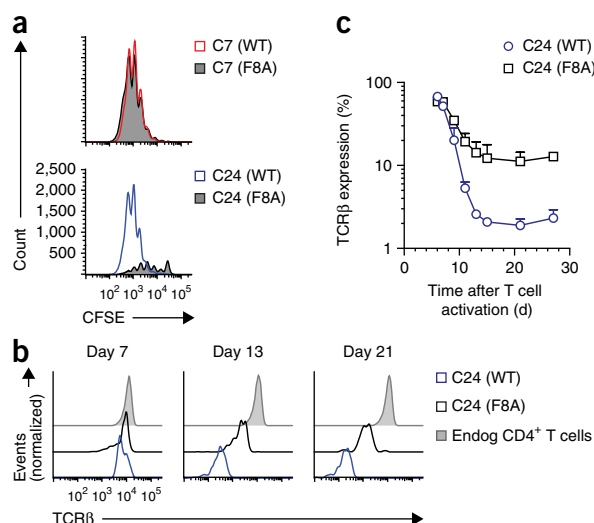


Figure 8 Programmed TCR downregulation is driven by the strength of initial antigen recognition. **(a)** Flow cytometry analyzing the CFSE dilution of C7 T cells (top) and C24 T cells (bottom) activated for 4 d with wild-type ESAT6(1–20) (WT) or F8A mutant peptide (F8A) (5 μ g/ml each). **(b,c)** Flow cytometry analyzing TCR β expression by C24 T cells and endogenous (CD90.1⁻) CD4⁺ T cells in blood from mice given 1×10^6 CD90.1⁺ C24 T_H1 cells activated *in vitro* as in **a**, assessed on days 7, 13 and 21 after activation (above plots) (results normalized to the mode) **(b)**, and analysis over time of the data obtained as in **b**, relative to results obtained for endogenous CD4⁺ T cells **(c)**. Data are from one experiment representative of two independent experiments (mean + s.d. of $n = 5$ male mice per group in **c**).

in the response. In this context, we emphasize that the programmed TCR downregulation did not commence until clonal expansion had peaked, at which point there can be millions of T cells of very high avidity distributed over the host. It is conceivable that a few million of such cells with full TCR expression would be difficult to restrain, especially given that at the peak of clonal expansion, antigen might still be abundant²⁷. Therefore, these cells could generate substantial inflammatory damage if left unchecked⁴³. From this perspective, programmed TCR downregulation seems to be an elegant solution whereby the biological activity of a given T cell clone is 'tuned' according to its intrinsic antigen reactivity, which ultimately favors cells of moderate avidity.

It remains puzzling how T cells use the strength of initial antigen recognition to adjust TCR expression following days of clonal expansion. Published studies have shown that transient TCR downregulation during initial T cell activation also correlates with antigen avidity, which means that functionally more potent ligands induce greater TCR downregulation^{40,41}. In those experiments, TCR loss paralleled the extent of full TCR ζ phosphorylation and activation of the signaling kinase ZAP-70. Ultimately, such biochemical changes would have to induce chromatin remodeling, such that graded expression of a 'TCR-targeting factor', such as a ubiquitin ligase^{34–36}, is achieved once T cells reach full clonal expansion. Thus, it might be expected that other molecules that influence TCR signaling would also contribute to programmed TCR downregulation^{39,44}. Indeed, CD4⁺ T cells with very low expression of CD5 also display TCR downregulation following clonal expansion⁴⁵. In our studies, C24 T cells displayed lower expression of CD5 but greater basal phosphorylation of TCR ζ than that of C7 T cells, which indicated that both these factors could have added to the extensive loss of TCR by C24 T cells. These results also emphasize that while CD5 expression and basal phosphorylation of TCR ζ often correlate^{46,47}, there are exceptions to this. Since CD5 expression can also be upregulated on activated T cells⁴⁵, some care must be taken when interpreting studies of this marker.

Our findings have implications for the design of T cell-based immunization strategies, as it is now clear that T cells with the highest long-term functionality are those that recognize antigen with intermediate avidity rather than very high avidity^{10,48–50}. Given the effect of programmed TCR downregulation on T cell biological activity, monitoring of TCR expression should prove a useful biomarker for helping new therapies select the optimal repertoire of responding T cells.

METHODS

Methods and any associated references are available in the [online version of the paper](#).

Note: Any Supplementary Information and Source Data files are available in the [online version of the paper](#).

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AUTHOR CONTRIBUTIONS

A.M.G., M.S.G., E.G.P. and J.W.J.v.H. designed the study; A.M.G., H.X. and J.W.J.v.H. performed experiments and analyzed data; A.M.G. generated the C7 and C24 mice and the *L. monocytogenes*–ESAT6 strain; I.M.L. and B.S. bred and genotyped the C7 and C24 mice and provided technical assistance; J.W.J.v.H. first identified programmed TCR downregulation; and E.G.P. and J.W.J.v.H. wrote the manuscript.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Generation of C24 mice. C24 mice with transgenic expression of a TCR specific for the I-A^b-ESAT6(1–20) antigen of *M. tuberculosis* were generated on the C57BL/6J (B6) background as previously described for the C7 mice²². The transgene encoding the C24 TCR is driven by the human CD2 promoter and expresses the TCR α chain variable region 11.3 and joining region 32 (V α 11.3–J α 32) and TCR β chain variable region 4, diversity region 2.1 and joining region 2.1 (V β 4–D β 2.1–J β 2.1).

Mouse models. B6, B6.PL-*Thy1^a/CyJ* (B6.PL), B6.129S7-*Rag1^{tm1Mom}/J* (*Rag1^{-/-}*) and B6.129S2-H2^{dIAb1-Ea/J} (MHC II^{-/-}) mice were purchased from the Jackson Laboratory. All animal procedures were performed in agreement with approved protocols by the Memorial Sloan-Kettering Cancer Center Institutional Animal Care and Use Committee. C7 and C24 mice were bred to B6.PL and *Rag1^{-/-}* mice (which lack mature B cells and T cells) to generate *Rag1^{-/-}* C7.PL and *Rag1^{-/-}* C24.PL mice, which were used as CD4⁺ T cell donors in all experiments. C7 and C24 mice were maintained in the heterozygous state for the TCR-encoding transgenes. All animals were used between 6 and 12 weeks of age, were maintained under specific pathogen-free conditions in individually ventilated cages, were kept on a 12 h light-dark cycle and had access to food and water ad libitum. The sizes of experimental groups were determined by the investigators without the use of statistical methods. All age- and sex-matched mice of a given strain were considered genetically identical and were randomly assigned to treatment groups, without blinding of the investigator to mouse identity. Both male and female mice were used throughout the study; however, T cell donors and recipients were always sex matched. In case T cell recipients showed clear signs of graft rejection (>100-fold lower T cell frequency compared to any other mouse in the same experimental group), these samples were excluded from further analysis.

MHC class II tetramer and antibody staining. For MHC class II tetramer staining, cells were incubated with various concentrations of phycoerythrin (PE)-labeled I-A^b-ESAT6(1–20) tetramers (NIH Tetramer Core Facility), in RPMI medium containing L-glutamine, 25 mM HEPES, 10% FBS and 200 μ M 2-mercaptoethanol (all Invitrogen) for 1 h at 37 °C. For tetramer dissociation, cells were incubated as above but with the addition of 0.1% NaN₃ (Sigma-Aldrich), to prevent tetramer internalization, and dissociation was followed after washing free tetramer away. For isolation of tetramer-binding cells, stained cells were purified using anti-PE MicroBeads and LS columns (both Miltenyi Biotec). For restimulation, cells were incubated in RPMI medium containing GolgiPlug (BD Biosciences) and 5 μ g mL⁻¹ ESAT6(1–20) peptide or 50 ng mL⁻¹ phorbol 12-myristate 13-acetate (PMA; Calbiochem) and 1 μ g mL⁻¹ Ionomycin (Sigma-Aldrich) for 5 h at 37 °C. For cell surface staining, cells were incubated in PBS containing 0.5% bovine serum albumin (BSA; Sigma-Aldrich) and 0.1% NaN₃ for 15 min at 4 °C. For intracellular staining, cells were fixed and permeabilized after cell surface staining in Cytofix/Cytoperm solution (BD Biosciences) for 15 min at 4 °C, followed by blocking in Perm/Wash solution (BD Biosciences) for 15 min at 4 °C and staining for intracellular proteins in Perm/Wash solution for 15 min at 4 °C. For nuclear staining, cells were stained using a Foxp3 Staining Buffer Set (eBioscience). Antibodies to the following proteins were used (clone numbers and dilutions in parentheses; **Supplementary Table 1**): CD3 ϵ (145-2C11; 1:100), CD4 (RM4-5; 1:100), CD5 (53-7.3; 1:100), CD8 α (53-6.7; 1:50), CD25 (PC61; 1:100), CD27 (LG.3A10; 1:100), CD28 (37.51; 1:50), CD44 (IM7; 1:100), CD45 (30-F11; 1:200), CD62L (MEL-14; 1:100), CD69 (H1.2F3; 1:50), CD90.1 (OX-7; 1:1,600), CD122 (TM- β 1; 1:50), CTLA-4 (UC10-4F10-11; 1:50), IFN- γ (XMG1.2; 1:200), PD-1 (J43; 1:50), TCR β (H57-597; 1:100), hamster IgG1 isotype control (A19-3; 1:50) and rat IgG2a isotype control (R35-95; 1:100) (all BD Biosciences); CD127 (A7R34; 1:50) and Foxp3 (FJK-16s; 1:100) (both eBioscience); and TCR ζ (H146-968; 1:100) (Thermo Scientific). Dead cells were excluded using 7-AAD Viability Staining Solution (eBioscience). Cells were measured on an LSRII flow cytometer (BD Biosciences) and data were analyzed using FlowJo software (TreeStar).

In vitro T_H1 differentiation and adoptive cell transfer. For *in vitro* T_H1 differentiation of C7 and C24 T cells, CD4⁺ T cells from pooled spleen and lymph nodes of donor mice were purified using CD4 (L3T4) MicroBeads and

LS columns (both Miltenyi Biotec). To prepare antigen-presenting cells, B6 spleens were depleted of endogenous T cells using CD90.2 MicroBeads and LS columns, after which remaining splenocytes were irradiated with 25 Gy. 4 $\times$ 10⁶ CD4⁺ T cells were co-cultured with 12 $\times$ 10⁶ irradiated antigen-presenting cells in RPMI medium containing 5 μ g mL⁻¹ ESAT6(1–20) peptide, 10 ng mL⁻¹ recombinant mouse IL-12 (R&D Systems) and 5 μ g mL⁻¹ neutralizing antibody to mouse IL-4 (11B11 hybridoma). Cells were cultured in 6-well plates at 37 °C and 5% CO₂. On days 2 and 3 of culture, cells were split in half and 12.5 ng mL⁻¹ recombinant mouse IL-2 (R&D Systems) was added. On day 4 of culture, C7 and C24 T_H1 cells were harvested, washed in PBS and transferred intravenously into sex-matched B6 recipients.

For *in vitro* cell proliferation assays, purified C7 and C24 T cells were resuspended in PBS containing 0.1% BSA and labeled with 5 μ M CFSE (Invitrogen) for 10 min at 37 °C. After washing in RPMI medium, cells were activated using antigen-presenting cells as described above. The percentage divided cells was calculated using the Proliferation tool of FlowJo software.

For adoptive transfer of naive C7 and C24 T cells, CD4⁺ T cells were purified from pooled spleen and lymph nodes of donor mice using the CD4⁺ T cell Isolation Kit II (Miltenyi Biotec) and LS columns, to generate untouched CD4⁺ T cells. Following washing in PBS, 1 $\times$ 10⁴ CD4⁺ T cells were transferred intravenously into sex-matched B6 recipients.

***M. tuberculosis* infection and T cell analysis.** *M. tuberculosis* Erdman was grown in Middlebrook 7H9 broth (BD Biosciences) until early log phase (OD₆₀₀ of ~0.2). Cultures were washed in PBS containing 0.05% Tween 80 (Fischer Scientific), sonicated and resuspended in water. B6 mice were infected with aerosolized bacteria using an inhalation exposure system (Glas-Col), which was calibrated to deliver ~100 colony-forming units (CFU) per animal.

To monitor C7 and C24 T cell responses over time, blood samples were harvested in PBS containing 5 mM EDTA (US Biological), followed by red blood cell lysis using NH₄Cl buffer. To determine lung T cell responses, left lungs were minced using scissors and digested in PBS containing 2% FBS, 500 U mL⁻¹ Collagenase Type 4 (Worthington) and 20 μ g mL⁻¹ DNase I (Roche) for 30 min at 37 °C. Cell suspensions were homogenized by passage through 18G needles followed by 100 μ M cell strainers (BD Biosciences). Blood and lung samples were stained as described above, fixed in PBS containing 2% paraformaldehyde (Electron Microscopy Sciences) and analyzed by flow cytometry.

To determine lung bacterial burdens, right lungs were homogenized in PBS containing 0.05% Tween 80, after which serial dilutions were made in PBS containing 0.05% Tween 80. Serial dilutions were spread on Middlebrook 7H10 agar (BD Biosciences) and incubated for 21 d at 37 °C and 5% CO₂, after which the number of colonies was determined.

Immunoblot analysis. Naive C7 and C24 T cells were purified from spleens of donor mice using CD4 MicroBeads and LS columns. 1 $\times$ 10⁶ cells were lysed in 100 μ L RIPA buffer (Sigma-Aldrich) containing protease and phosphatase inhibitors (Thermo Scientific). 15 μ L of lysates was mixed with 2 $\times$ loading buffer, separated on 10% SDS-PAGE and transferred to PVDF membrane. Membranes were incubated with the primary antibodies rabbit antibody to TCR ζ phosphorylated at Tyr142 (SAB4200334) and rabbit anti- β -actin (SAB5500001) (both Sigma-Aldrich) overnight at 4 °C, with a dilution factor of 1:1,000 in blocking buffer (PBS with 5% skim milk and 0.1% Tween 20). Following incubation with horseradish peroxidase (HRP)-linked anti-rabbit secondary antibody (7074; Cell Signaling; diluted 1:5,000 in blocking buffer), detection was performed using ECL reagent (GE Healthcare). To allow fair comparison, blots containing staining for phosphorylated TCR ζ and β -actin were imaged simultaneously. Densitometry was assessed using ImageJ software (NIH).

Recombinant *L. monocytogenes*-ESAT6 infection. To generate a recombinant *L. monocytogenes* strain that stably expresses and secretes ESAT6, the entire ESAT6-encoding region of *M. tuberculosis*, except the first ATG (ESAT6 (2–95)), was fused in-frame behind the *hly* signal sequence of the gene encoding *L. monocytogenes* secreted protein listeriolysin O. The *hly*-ESAT6 (2–95) fusion product was cloned behind the *hly* promoter and the entire construct was inserted into the pPL2 shuttle integration vector³². The pPL2-ESAT6

vector was then introduced as a single copy into the genome of the *L. monocytogenes* strain 10403S via conjugation. Following selection by chloramphenicol, successful transconjugants were confirmed by PCR and passaged through B6 mice to ensure virulence.

For recombinant *L. monocytogenes*–ESAT6 infection, bacteria were grown in Brain Heart Infusion broth (BD Biosciences) until early log phase (OD_{600} of 0.1). Cultures were washed in PBS and injected intravenously into B6 mice at a dose of 5×10^3 CFU for primary infections and 5×10^5 CFU for secondary infections.

Real-time PCR analysis. To analyze gene expression of adoptively transferred CD90.1⁺ C7 and C24 T cells, these cells were purified from the spleens of CD90.2⁺ recipient mice using CD90.1 MicroBeads (Miltenyi Biotec) and separation over two consecutive MS columns (Miltenyi Biotec). Eluted cells

were lysed in RLT buffer (QIAGEN), homogenized using QIAshredder columns (QIAGEN) and stored at -80°C . Total RNA from frozen homogenates was extracted using an RNeasy mini kit (QIAGEN), followed by cDNA generation using a QuantiTect Reverse Transcription kit (QIAGEN). TaqMan probe-based real-time PCR analysis was performed on a StepOnePlus Real-Time PCR System (Applied Biosystems), using a DyNAmo probe qPCR kit (Thermo Scientific) and specific TaqMan probes (Invitrogen). *Actb* was used as endogenous control. Relative gene expression was determined using the Comparative C_T method.

Statistical analysis. Statistical analysis was performed using an unpaired two-tailed Student's *t* test, assuming that data was normally distributed and there was equal variance between the groups under comparison. $P < 0.05$ was considered statistically significant.